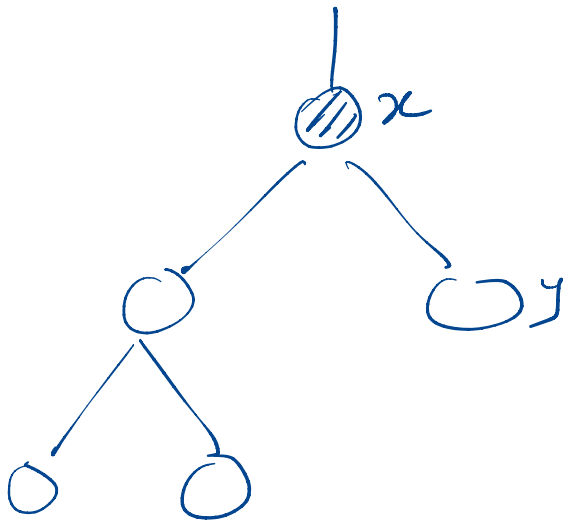


AVL Trees - II



$\text{Height}(x) = \# \text{ edges}$
in the subtree
rooted at x
along the longest path
until the leaf

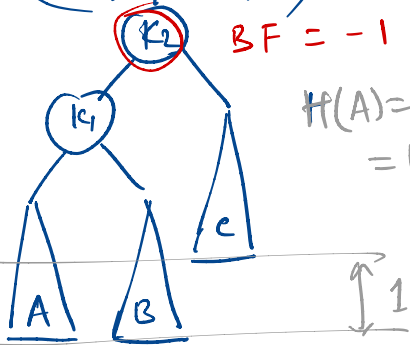
Balancing in AVL trees

AVL Add

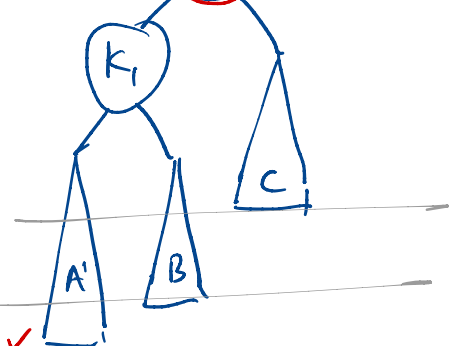
(Left-Left): LL

BF = -1

$$H(A) = H(B) = H(C)$$



BF = -2

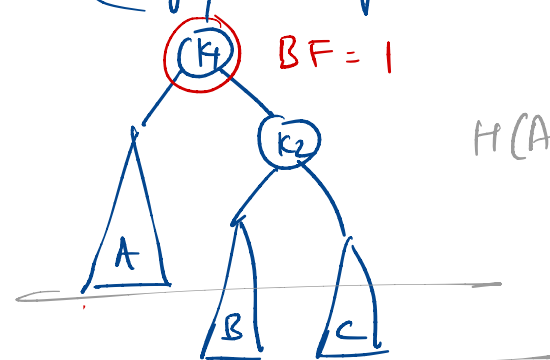


Cases :

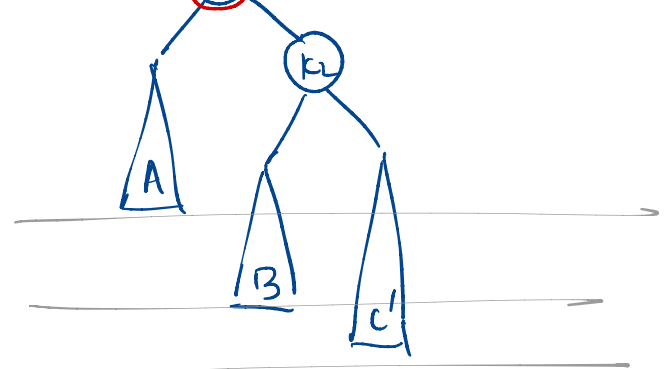
(Right-Right): RR

BF = 1

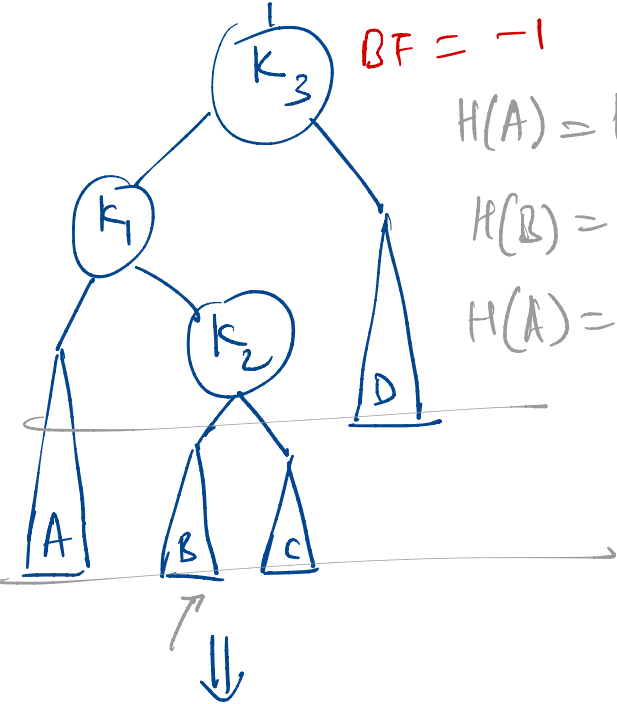
$$H(A) = H(B) = H(C)$$



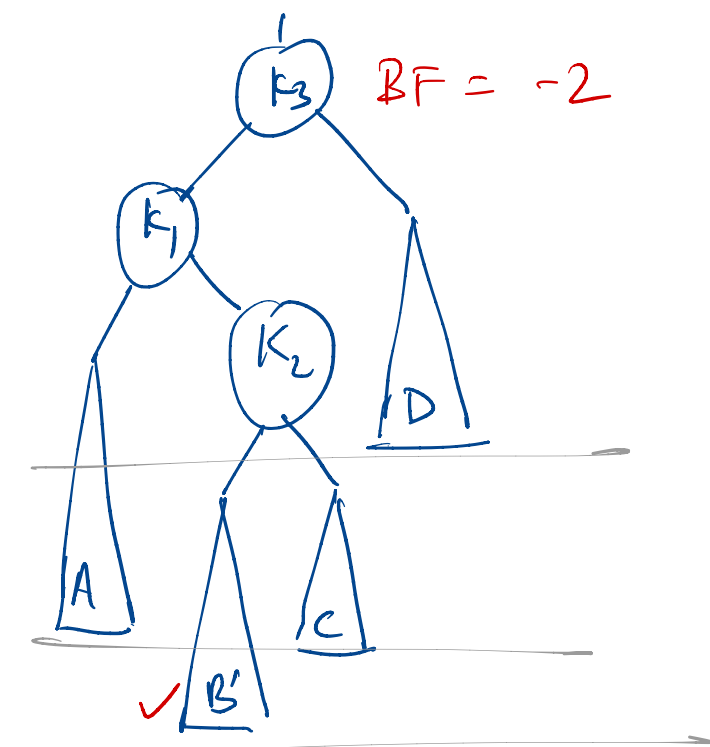
BF = 2



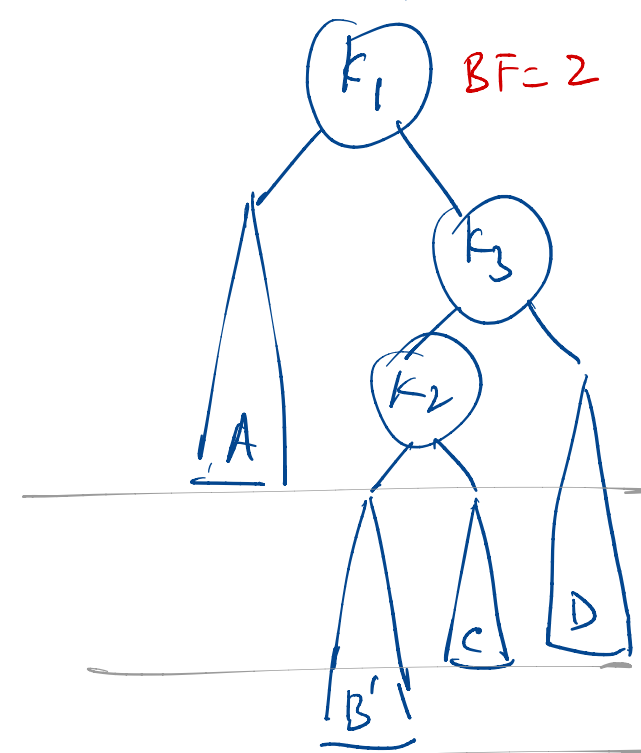
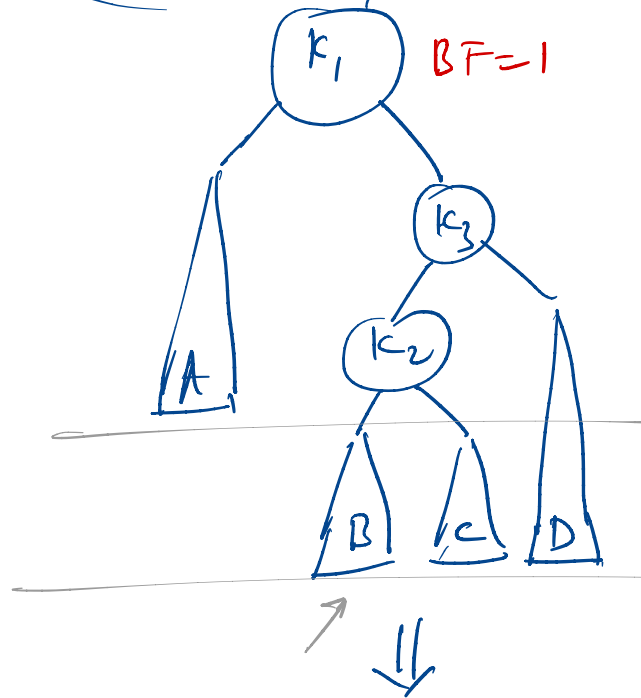
(Left-Right): LR



$$\begin{aligned} H(A) &= H(D) \\ H(B) &= H(C) \\ H(A) &= H(B) + 1 \end{aligned}$$

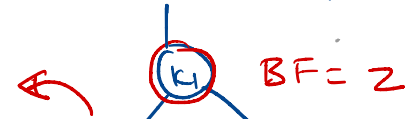


(Right-Left) RL

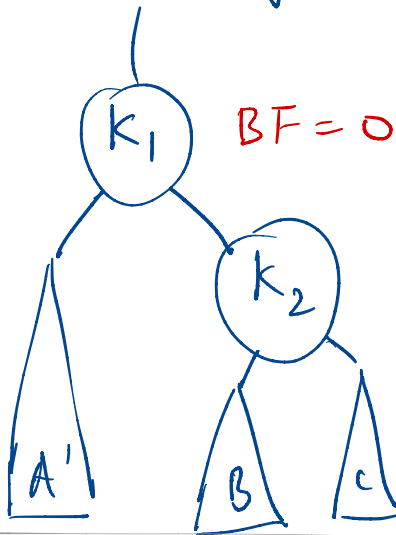


Rebalancing with rotations

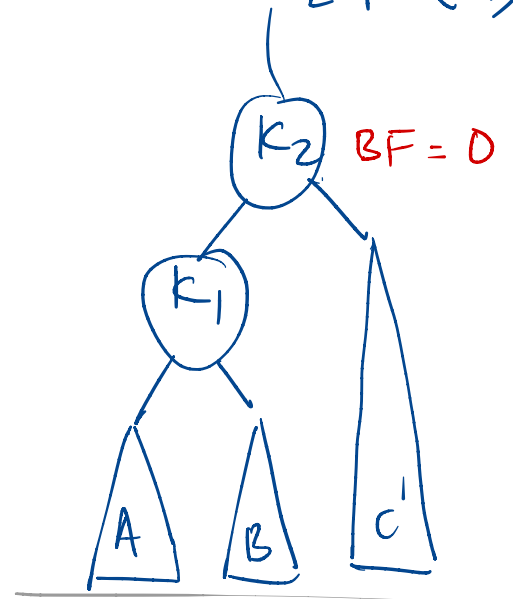
LL



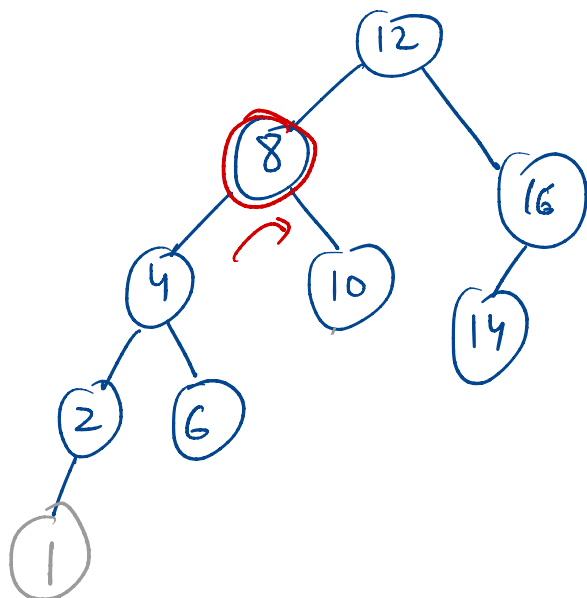
Rotate - Right (K_2)



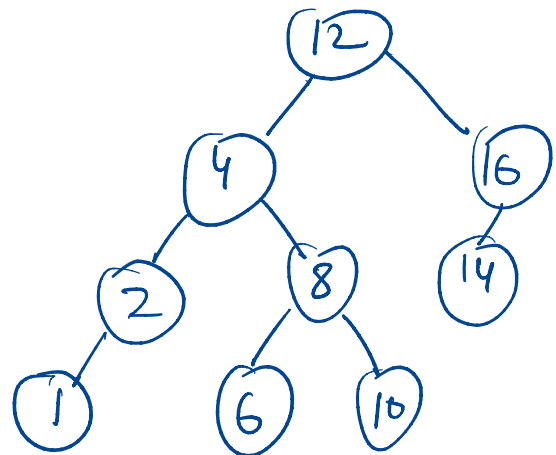
Rotate - Left (K_1)

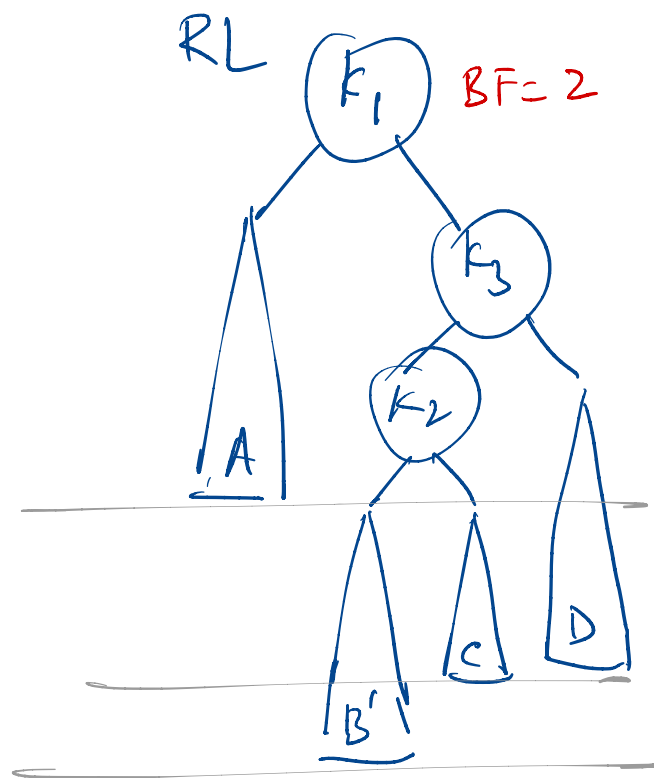
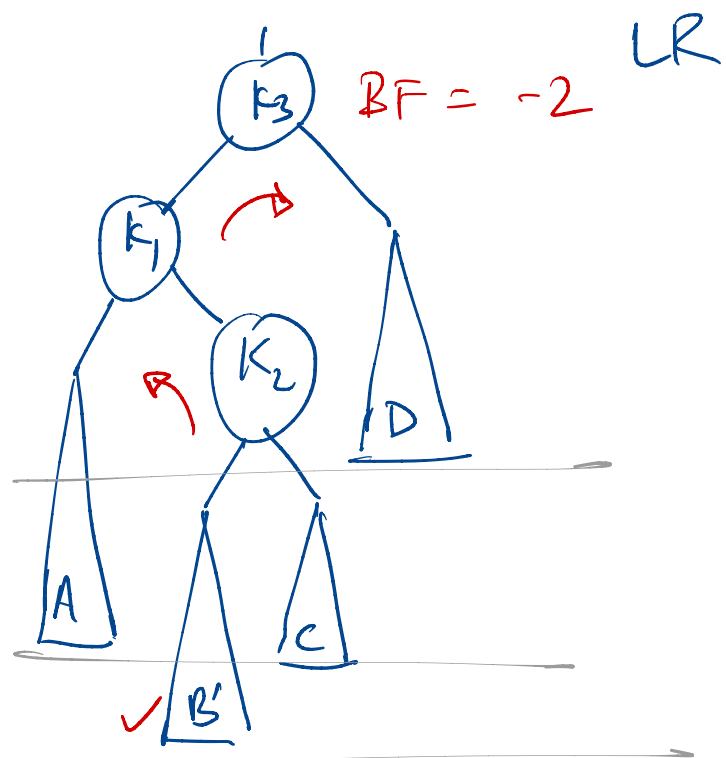


Example : Right rotation



$\Rightarrow$

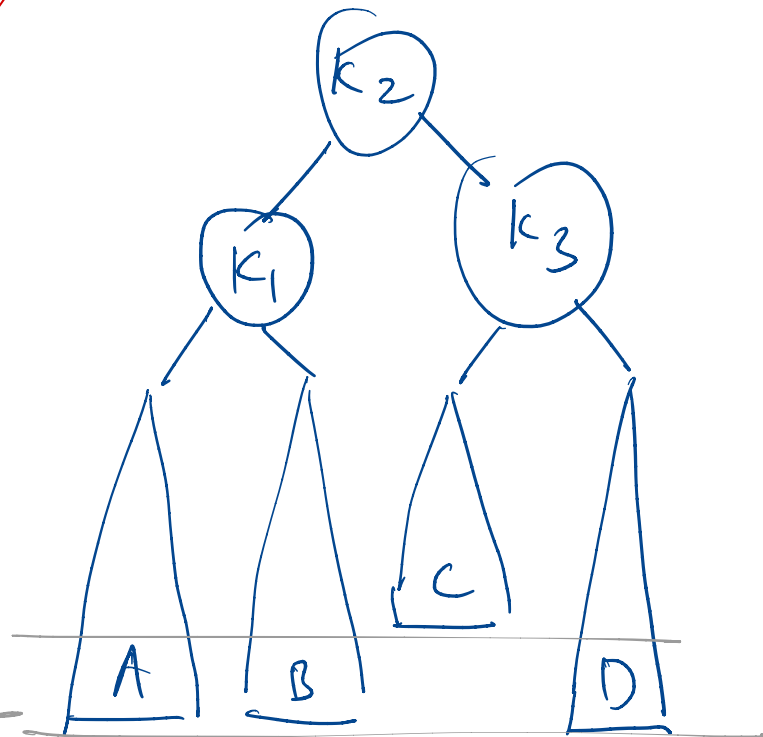
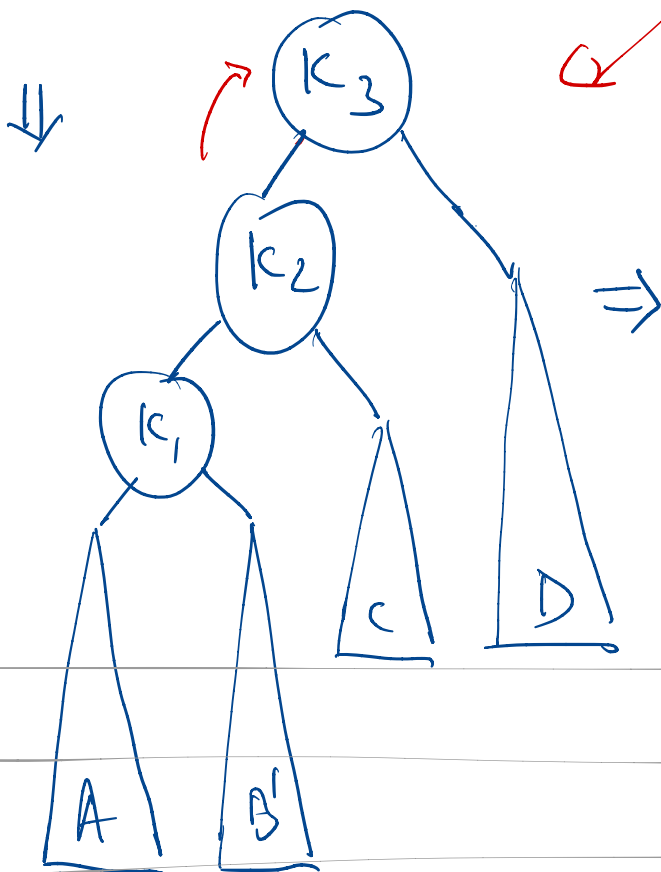




Two rotations are needed

To Do

- ① Left-Rotate (K_1)
- ② Right-Rotate (K_3)



Implementing add

- Perform normal BST add
- Update balance factors for each node
(only BF's of ancestors of newly added node will change)
- If a node's BF becomes -2 or $+2$, rebalance it.

Cases:

- Current node's BF < -1 where imbalance is detected

R rotation — if Left child $BF < 0 \Rightarrow$ LL case

LR rotation — if Left child $BF > 0 \Rightarrow$ LR case

- Current node's $BF > 1$

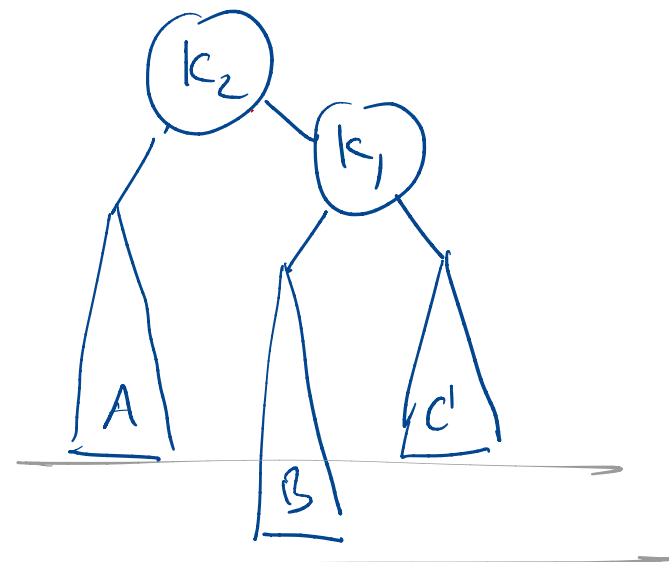
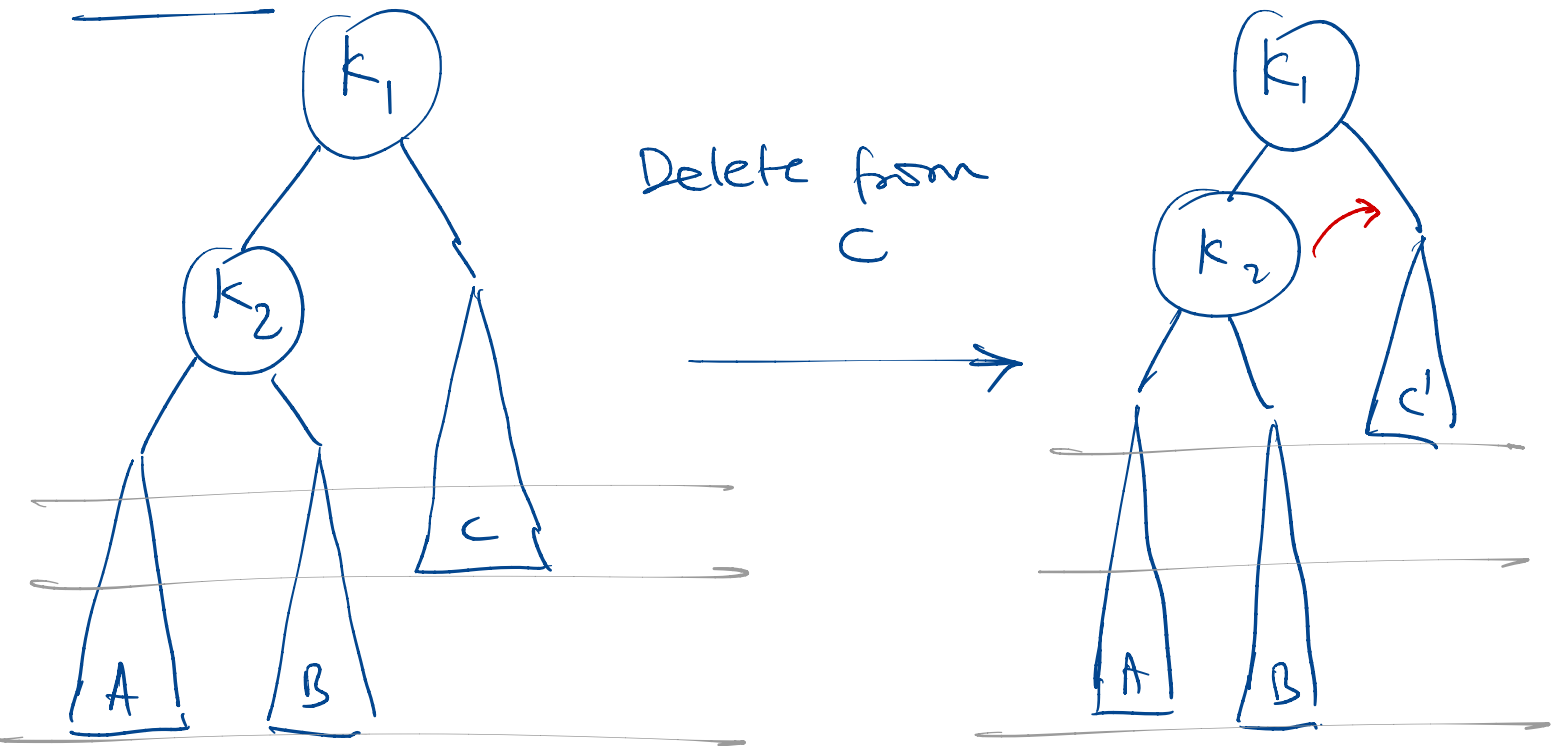
RL rotation — if right child $BF < 0 \Rightarrow$ RL case

L rotation — if right child $BF > 0 \Rightarrow$ RR case

AVL tree node deletion

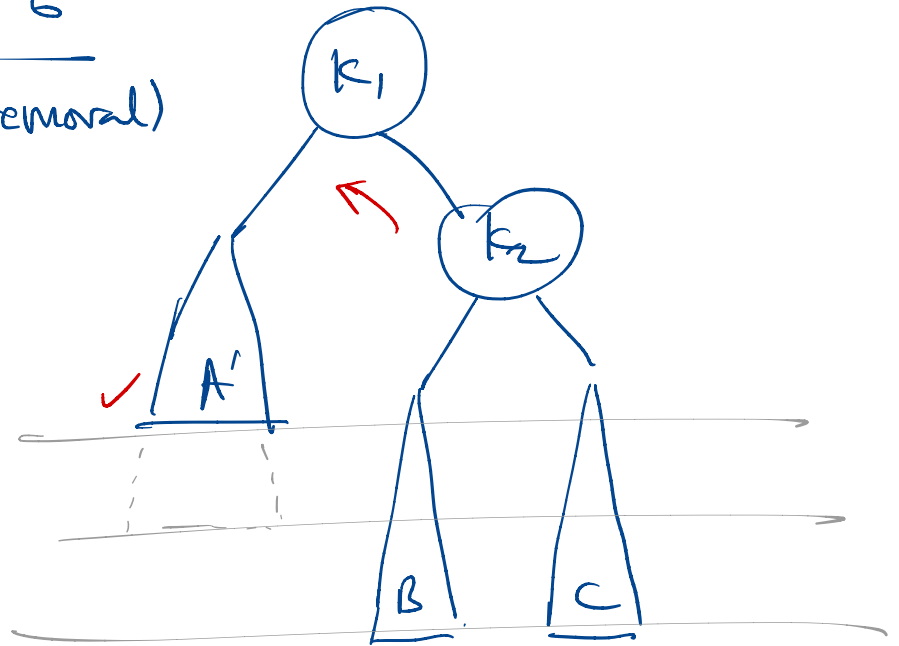
- Same 4 cases as in node add.
- Two additional cases that can be handled with a single rotation

Case 5

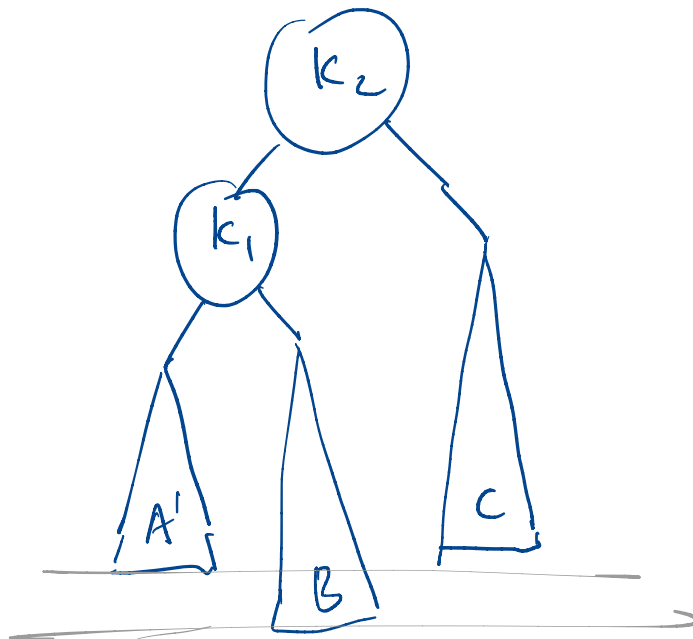


- neither LL nor LR
 - Can be fixed with a right rotation about k_1
- Rotate-Right (k_2)

Symmetric case 6
(after removal)



fix: Rotate - left (k_1)



Implementing Delete

- Perform normal BST Delete
- Update balance factors for each node
(only BF's of ancestors of newly added node will change)
- If a node's BF becomes -2 or $+2$, rebalance it.

Cases:

- Current node's BF < -1 where imbalance is detected

R rotation — if left child BF $< 0 \Rightarrow$ LL case

LR rotation — if left child BF $> 0 \Rightarrow$ LR case

R rotation — if left child BF $= 0 \Rightarrow$ Case 5

- Current node's BF > 1

RL rotation — if right child BF $< 0 \Rightarrow$ RL case

L rotation — if right child BF $> 0 \Rightarrow$ RR case

L rotation — if right child BF $= 0 \Rightarrow$ Case 6